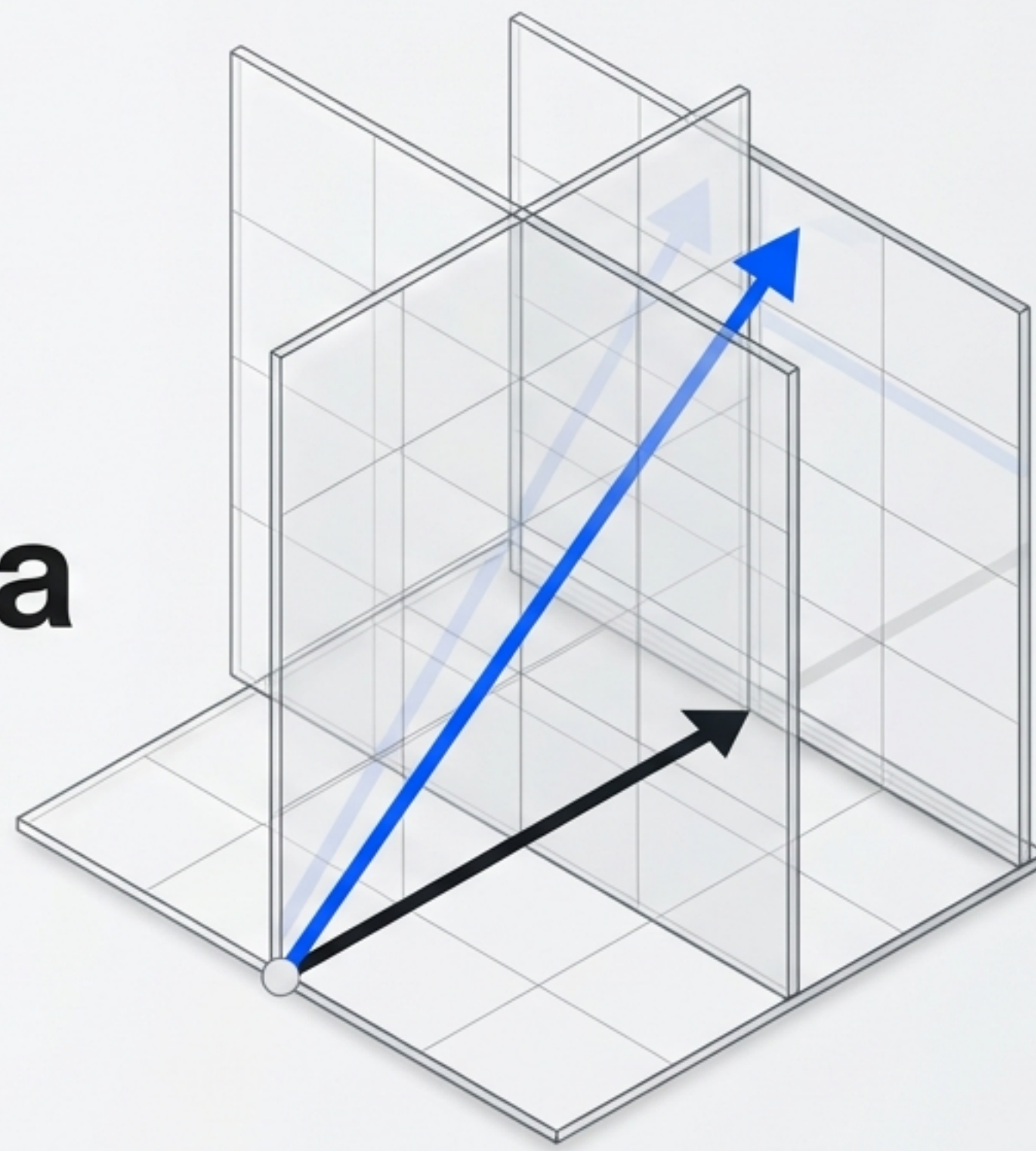


Architectural Precision

# Vector Mathematics: The Geometry of Data

Norms, Products, and Spatial Relationships



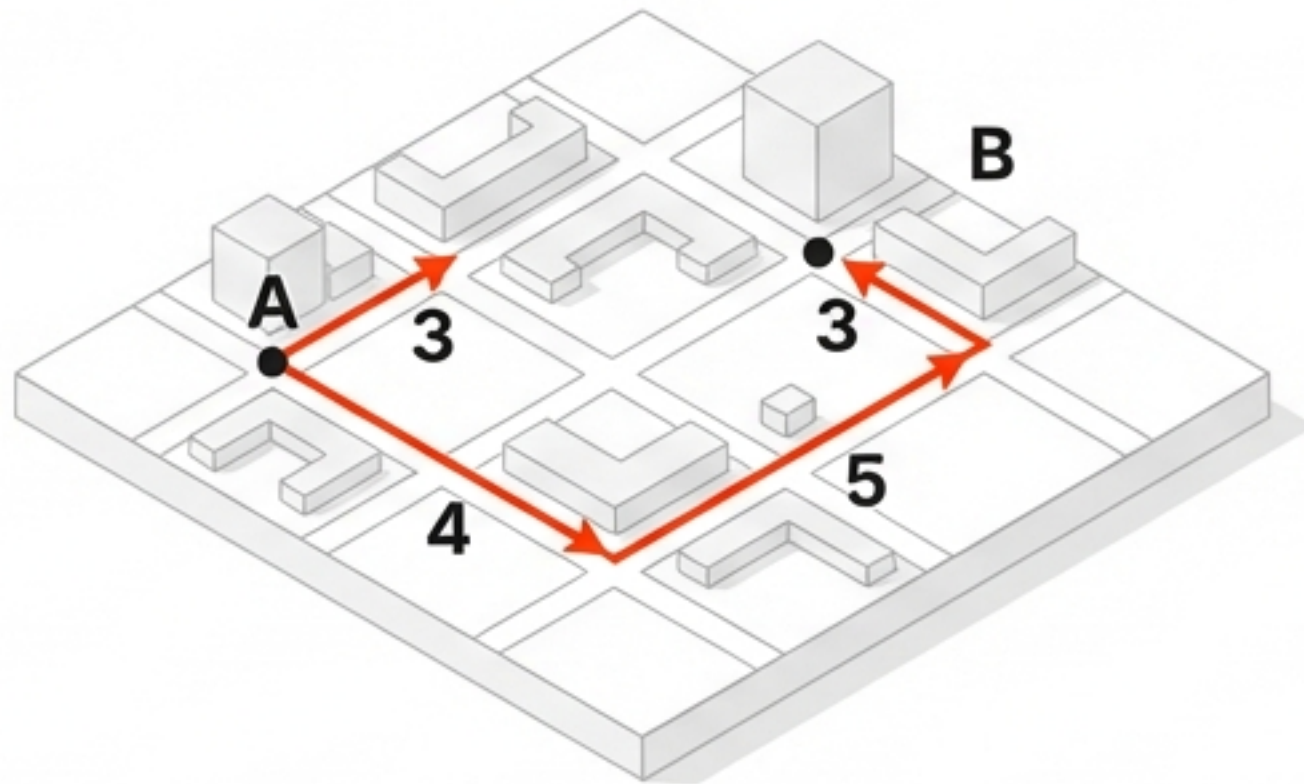


# How do we measure the size of a vector?

## The Manhattan Norm (L1)

Distance along a grid

$$\|\vec{x}\|_1 = \sum_{i=1}^n |x_i|$$



$$|3| + |-4| + |5| = \mathbf{12}$$

## The Euclidean Norm (L2)

Straight-line distance

$$\|\vec{x}\|_2 = \sqrt{\sum_{i=1}^n x_i^2}$$



$$\sqrt{3^2 + (-4)^2 + 5^2} = \sqrt{50} \approx \mathbf{7.07}$$

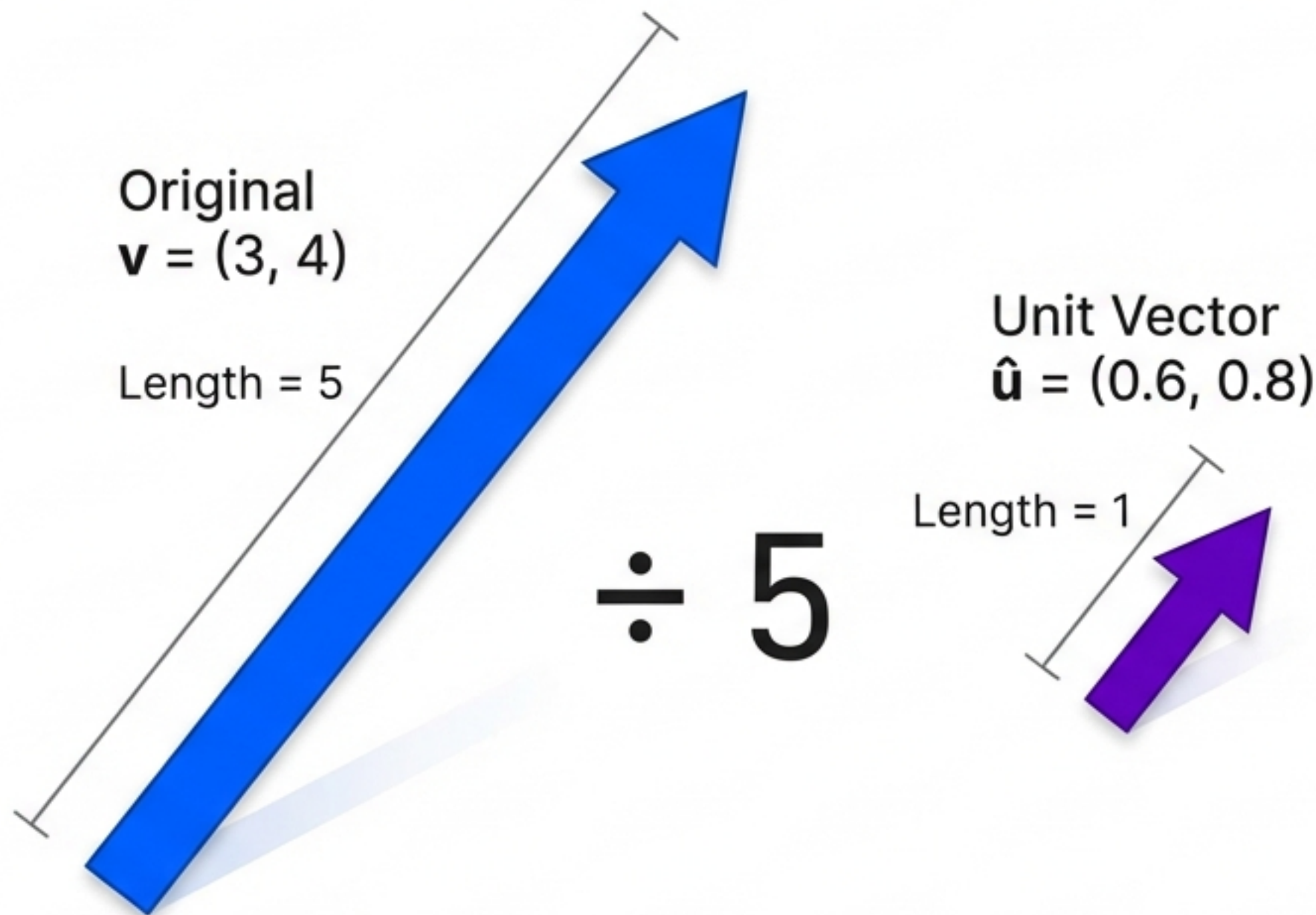
# Isolating direction via the Unit Vector

**Definition:** A vector with magnitude 1 that preserves the original direction.

**Notation:** Denoted by a “hat” symbol (e.g.,  $\hat{u}$ ).

**Formula:** The vector divided by its own magnitude.

$$\hat{u} = \frac{\mathbf{v}}{|\mathbf{v}|}$$

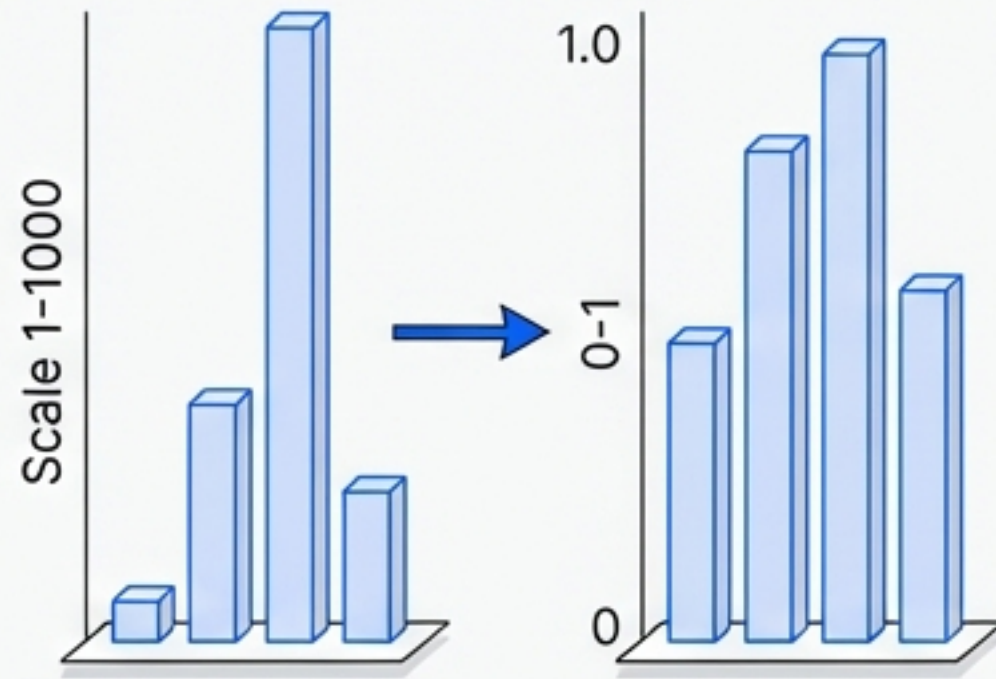


$$\sqrt{(0.6^2 + 0.8^2)} = \sqrt{(0.36 + 0.64)} = \sqrt{1} = 1$$



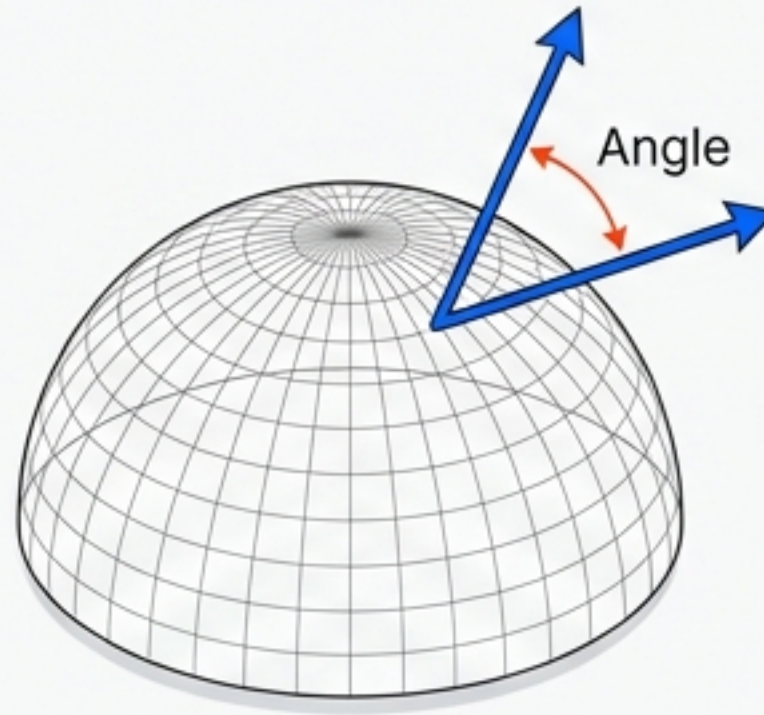
# Why Normalisation matters in Machine Learning

## Data Normalisation



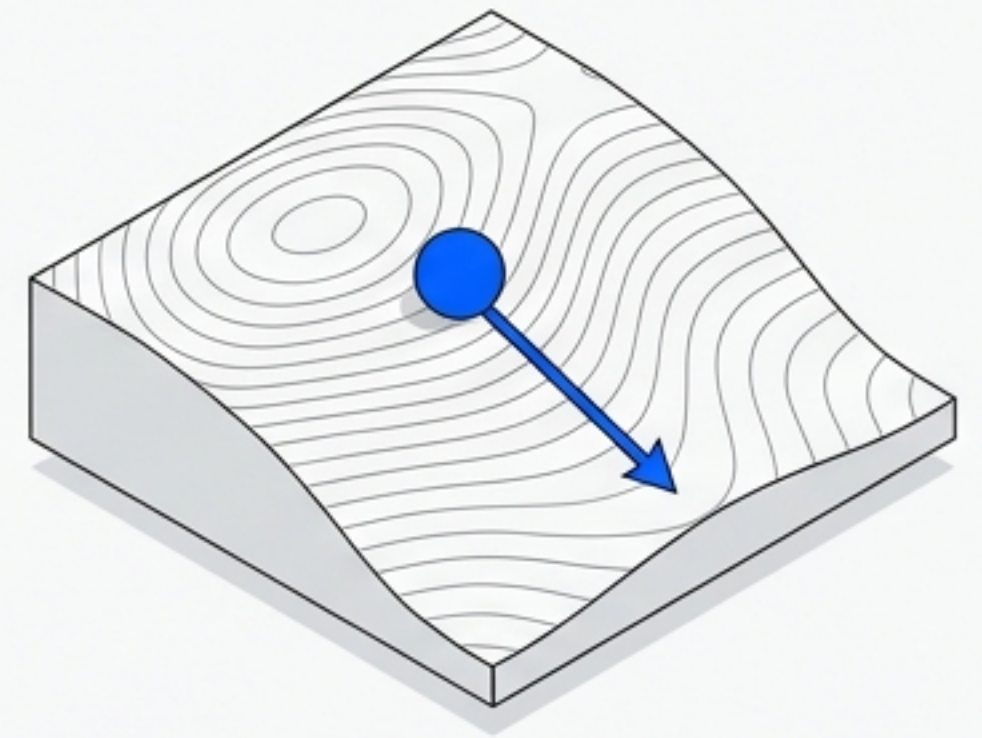
Scaling features for fair comparison.

## Cosine Similarity



Comparing orientation, ignoring magnitude.

## Gradient Descent



Finding the direction of steepest descent.



# The Dot Product measures similarity

This slide explains the concept of projection.

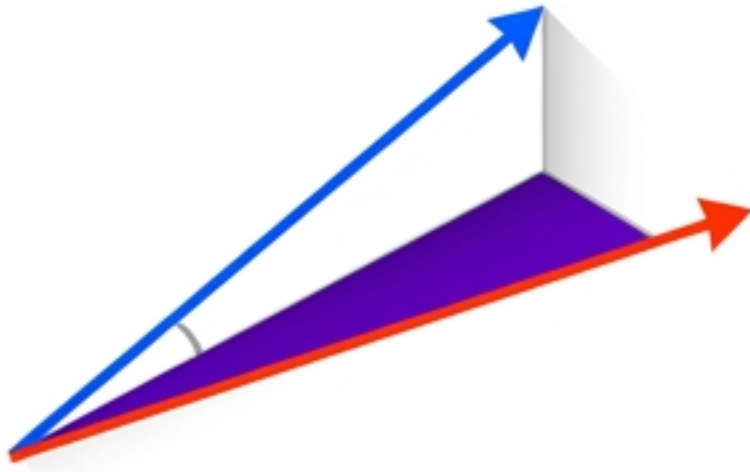
Algebraic

$$A \cdot B = \sum_{i=1}^n a_i * b_i$$

Geometric

$$A \cdot B = |A| |B| \cos \theta$$

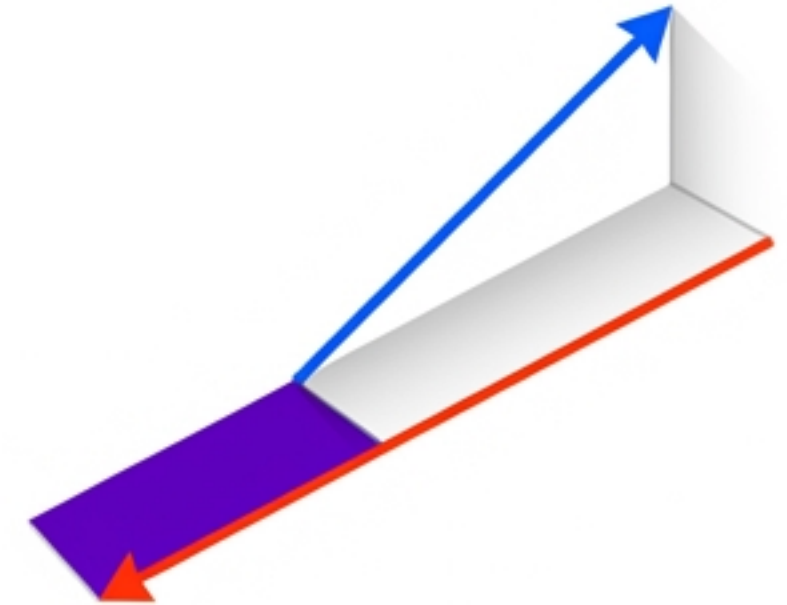
The Visual Intuition



**High Similarity**  
Result  $> 0$



**Orthogonal**  
Result  $= 0$



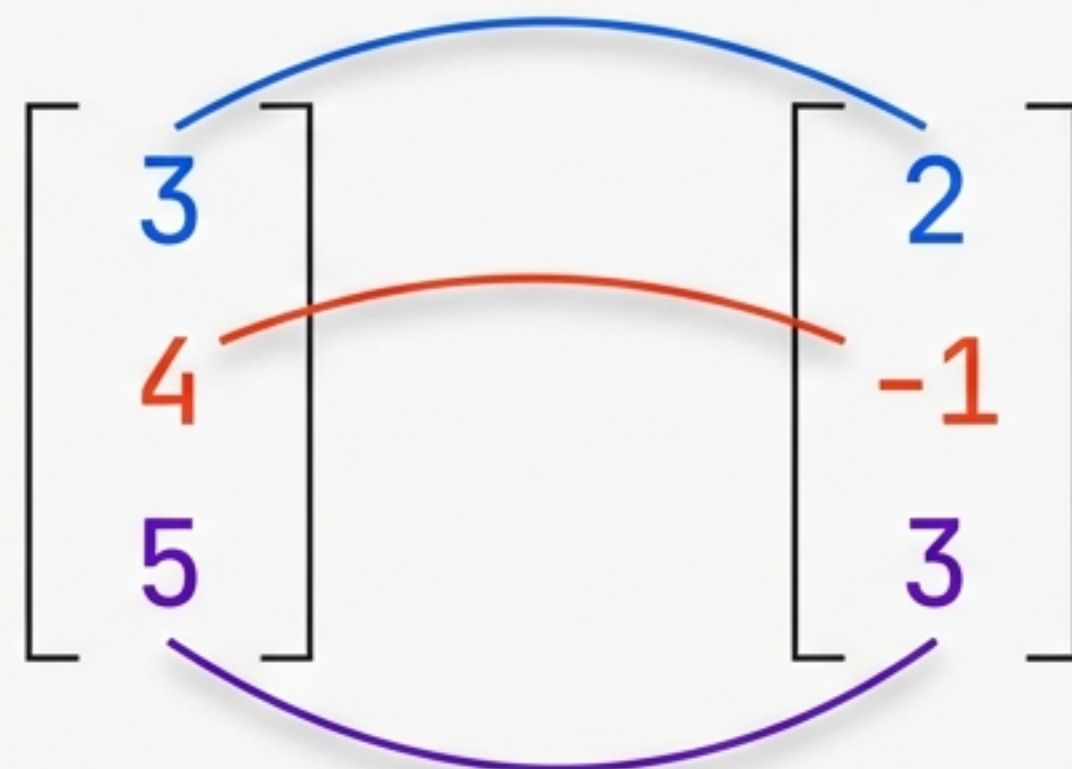
**Opposite**  
Result  $< 0$



# Calculating the Dot Product

This slide focuses on the mechanics of the calculation using the example vectors **A=(3,4,5)** and **B=(2,-1,3)**.

$$\begin{aligned} \mathbf{A} \bullet \mathbf{B} &= (3 \times 2) + (4 \times -1) + (5 \times 3) \\ &= 6 - 4 + 15 \\ &= 17 \end{aligned}$$



✓ So,  $\mathbf{A} \bullet \mathbf{B} = 17$

# Interpreting the Scalar Result

Parallel ( $\theta = 0^\circ$ ):

$$A \bullet B = |A||B| \text{ (Max Similarity)}$$

Perpendicular ( $\theta = 90^\circ$ ):

$$A \bullet B = 0 \text{ (Orthogonal)}$$

Opposite ( $\theta = 180^\circ$ ):

$$A \bullet B = -|A||B|$$

```
Terminal

import numpy as np

A = np.array([3, 4, 5])
B = np.array([2, -1, 3])

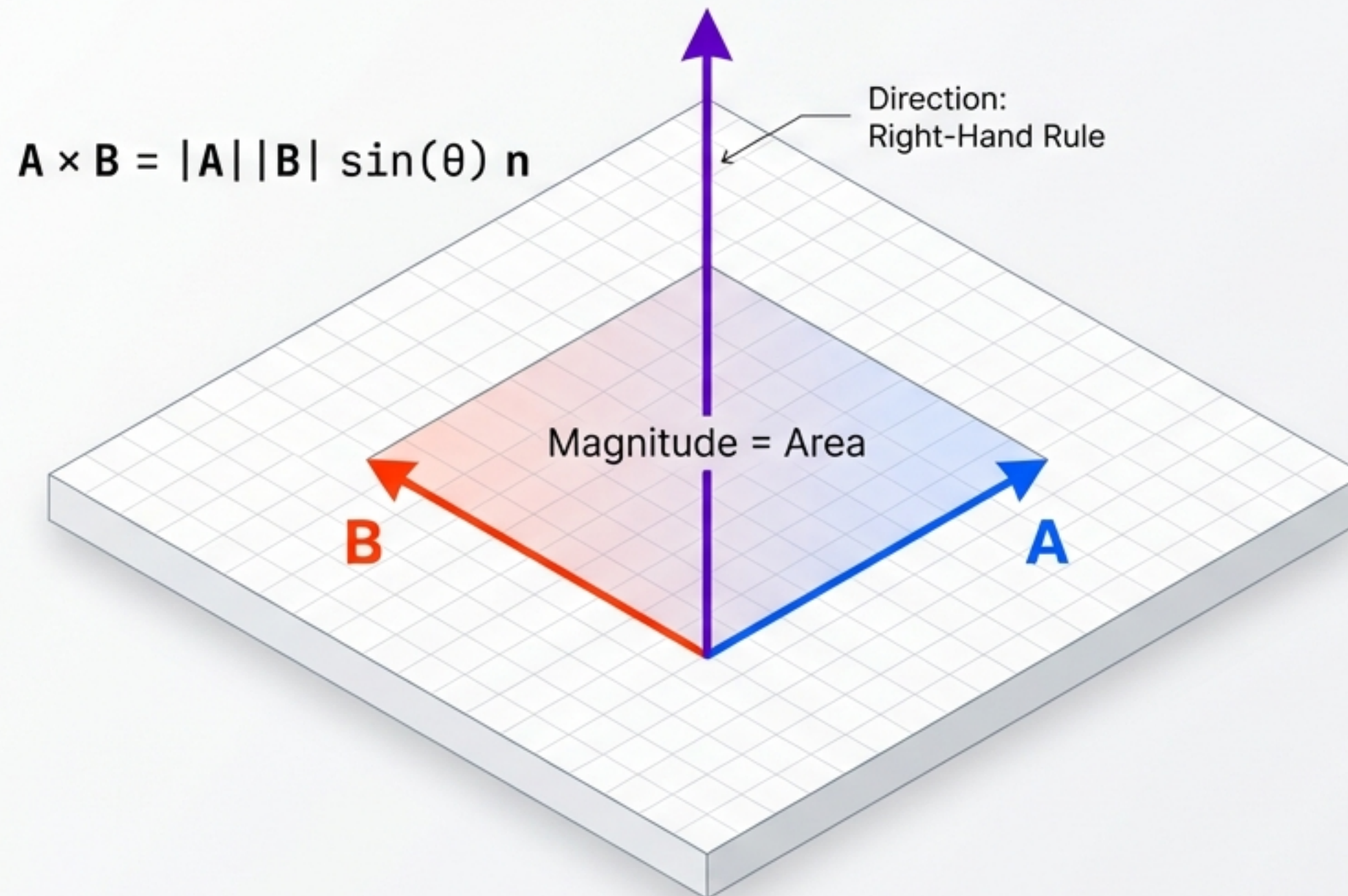
dot_product = np.dot(A, B)

print(f'Dot Product: {dot_product}')
```

> Dot Product: 17



# The Cross Product creates a new dimension





# Calculating via the Determinant

Step 1:

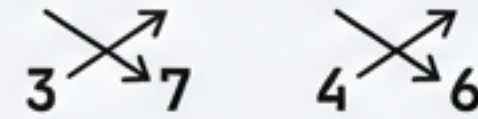
$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 3 & 4 \\ 5 & 6 & 7 \end{vmatrix}$$

Vector A Inter      Vector A Inter      Vector A Inter

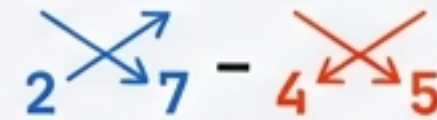
Vector B Inter      Vector B Inter      Vector B Inter

Step 2:

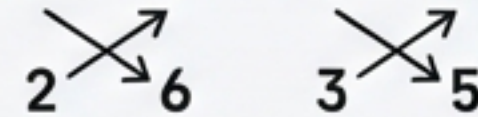
**i** component:  $(3 \times 7) - (4 \times 6) = 21 - 24 = -3$



**j** component:  $-[(2 \times 7) - (4 \times 5)] = -[14 - 20] = -[-6] = 6$



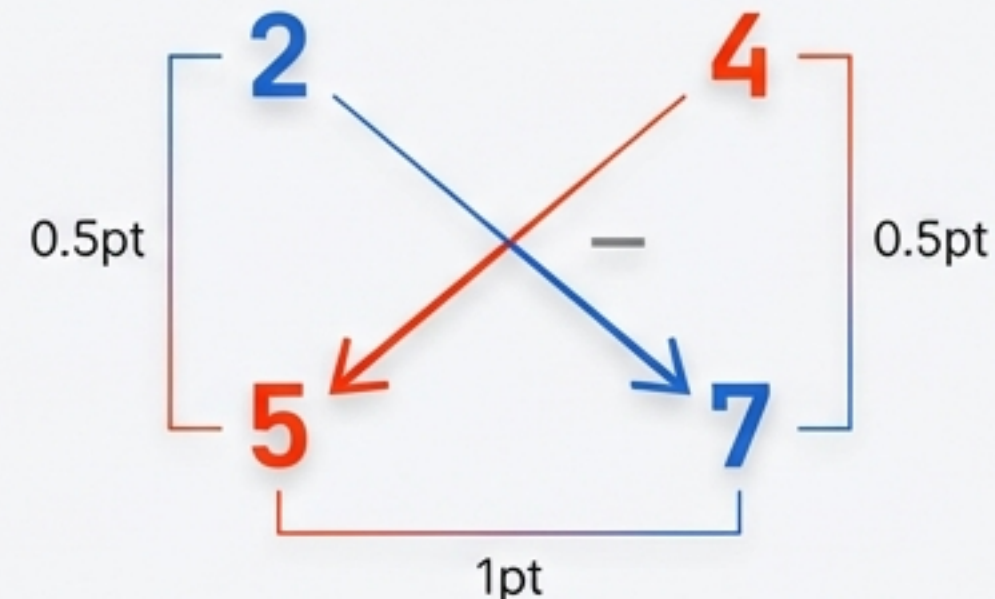
**k** component:  $(2 \times 6) - (3 \times 5) = 12 - 15 = -3$



Step 3:

**Result:  $(-3, 6, -3)$**

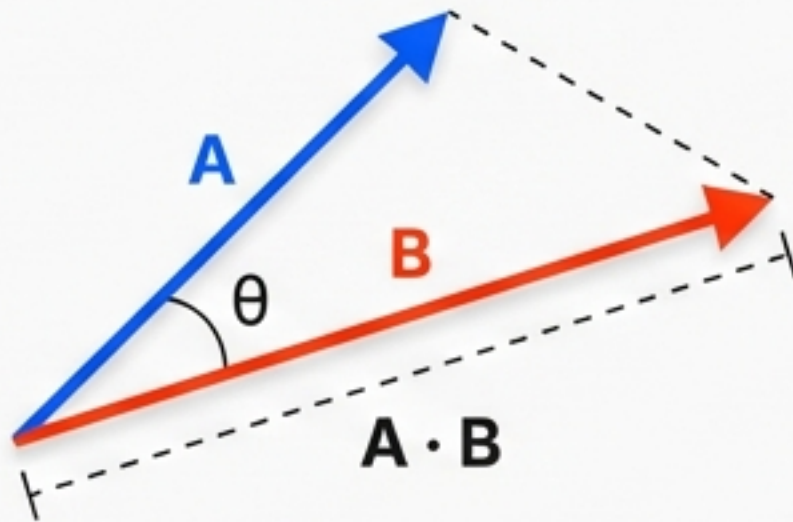
The resulting vector is orthogonal to both **A** and **B**.





# Dot vs. Cross: Choosing the right tool

## DOT PRODUCT



Result: SCALAR (Single Number)

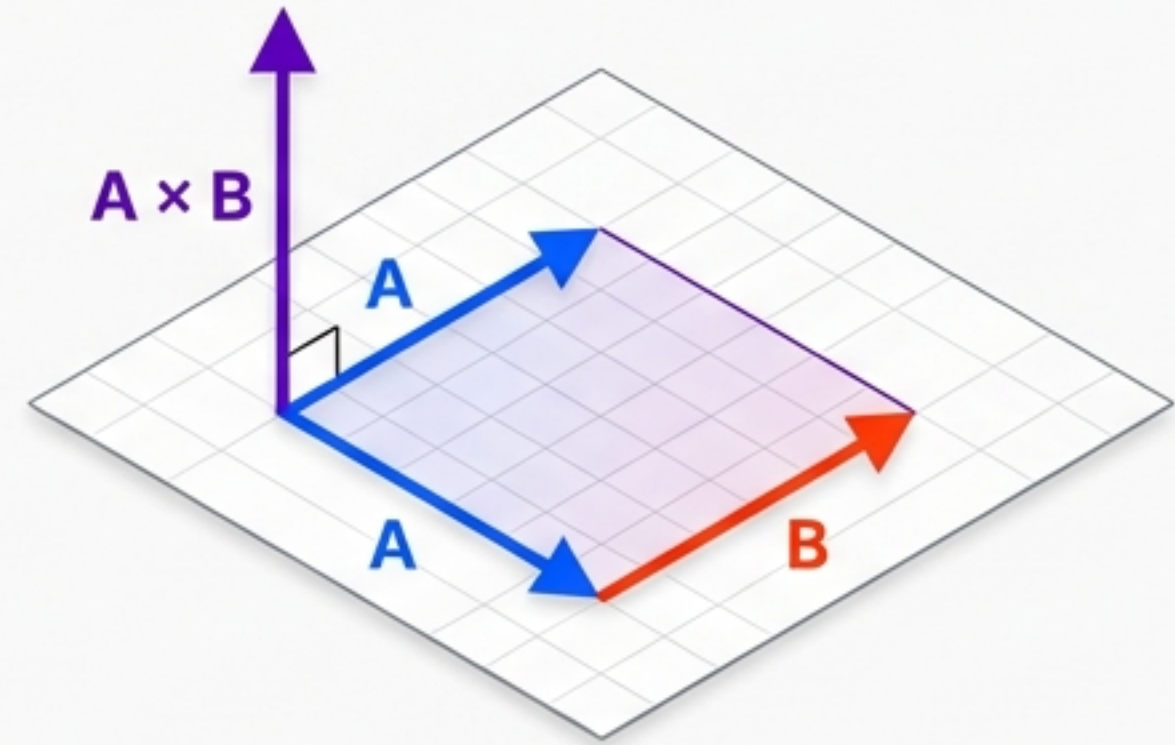
Measures: Similarity / Parallelism

Function: Cosine

Zero when: Perpendicular

Order:  $A \cdot B = B \cdot A$  (Commutative)

## CROSS PRODUCT



Result: VECTOR (Direction & Magnitude)

Measures: Area / Torque / Orthogonality

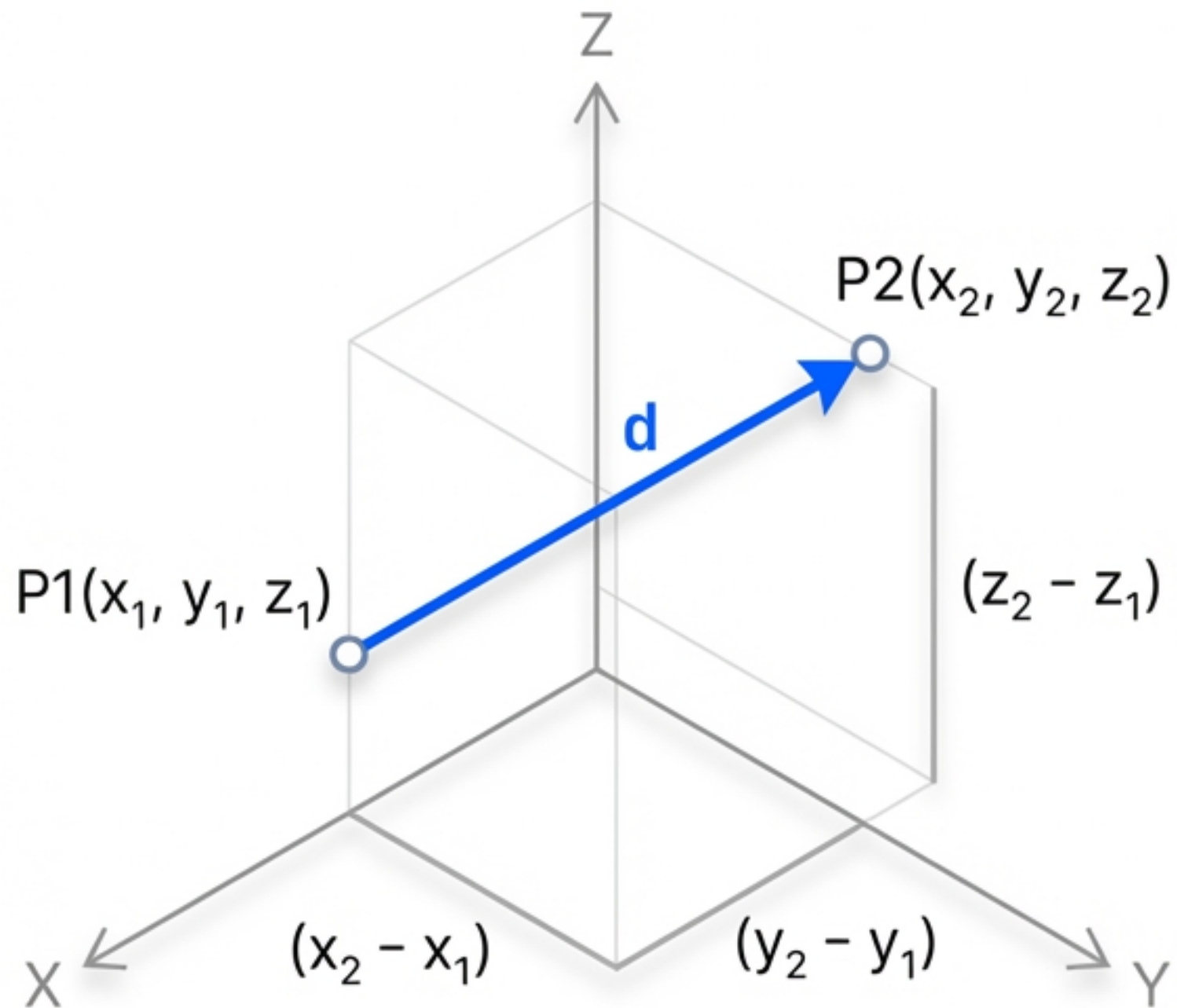
Function: Sine

Zero when: Parallel

Order:  $A \times B = -(B \times A)$  (Anti-commutative)



# Defining distance in vector space



Distance is simply the magnitude of the difference vector.

$$d(P_1, P_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$



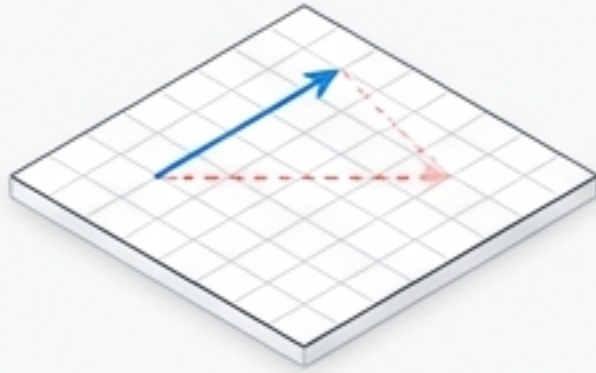
# Calculating spatial distance

## 2D DISTANCE

Points:

$$P_1(1, 2)$$

$$P_2(4, 6)$$



Difference:

$$d = P_2 - P_1 = (3, 4)$$

Calculation:

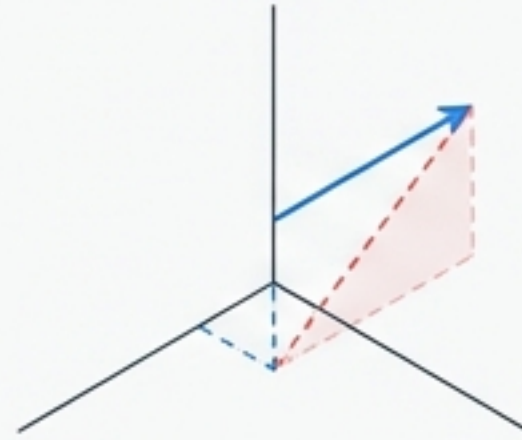
$$d = \sqrt{3^2 + 4^2} = 5 \text{ units} \checkmark$$

## 3D DISTANCE

Points:

$$P_1(2, -1, 3)$$

$$P_2(5, 2, 7)$$



Difference:

$$d = P_2 - P_1 = (3, 3, 4)$$

Calculation:

$$\begin{aligned} d &= \sqrt{3^2 + 3^2 + 4^2} \\ &= \sqrt{(9 + 9 + 16)} \\ &= \sqrt{34} \approx 5.83 \checkmark \end{aligned}$$

## DISTANCE FROM ORIGIN

Point:

$$P(x, y, z)$$

$$\text{Origin } O(0, 0, 0)$$



Distance:

Magnitude of Vector P

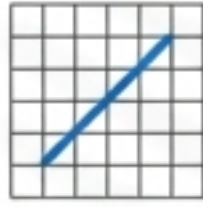
Calculation:

$$d = |P| = \sqrt{x^2 + y^2 + z^2}$$



# The Linear Algebra Reference Guide

**L1 Norm  
(Manhattan)**



$$\sum |x_i|$$

**L2 Norm  
(Euclidean)**



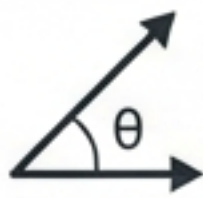
$$\sqrt{\sum x_i^2}$$

**Unit Vector**



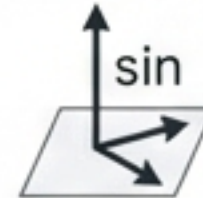
$$\frac{\mathbf{v}}{|\mathbf{v}|}$$

**Dot Product  
(Similarity)**



$$|\mathbf{A}| |\mathbf{B}| \cos \theta$$

**Cross Product  
(Area/Normal)**



$$|\mathbf{A}| |\mathbf{B}| \sin \theta \mathbf{n}$$

**Distance**



$$\sqrt{\Delta x^2 + \Delta y^2 + \Delta z^2}$$



# From Coordinates to Intelligence

These operations—measuring magnitude, calculating similarity, and determining orthogonality—are not just abstract math. They are the mechanisms we use to navigate 3D space, power recommendation engines, and render the graphics on this screen.

